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SERVICE BULLETIN
No. 113

ENAC Approval No. 2002-4194/SVNA dated 25/07/02

This document is the English translation of the ENAC approved Italian original.

M A N D A T O R Y

SUBJECT: *FUEL SYSTEM*

1. GENERAL

1.1 AIRCRAFT AFFECTED

All P68 Series aircraft.

1.2 PURPOSE

To supply more detailed instructions for the operational check of the fuel selector control system, scheduled for every 100 flight hours as per the ENAC Approved current Maintenance programme.

1.3 DESCRIPTION

This Service Bulletin has been prepared to address specific reports of malfunction or improper fuel selector valve control system rigging.

This bulletin has been issued to update the maintenance procedures of the fuel selector control system, to ensure its correct operation.

1. GENERAL (Cont.)

1.4 **COMPLIANCE**

Perform an initial inspection within the next 50 hours scheduled inspection and however not more than 3 months since receiving this bulletin. Then as required by the normal inspection programme, check the fuel selector valve control system for correct operation at 100 hour intervals and any time one or more of the following conditions become evident:

✂✂ Cabin fuel selector knob is stiff and difficult to operate

✂✂ The detent indicating the correct valve positions is no longer perceptible;

✂✂ Anomalous fuel level readings between tanks that cannot be attributed to different engine fuel consumption as indicated by the fuel flow gauge, or to evident fuel leakage;

✂✂ Uncommanded fuel transfer between tanks in cases where the aircraft has been parked on an unlevel surface. This condition may be noted by an evident imbalance of the aircraft (one wing low).

1.5 **TIME REQUIRED**

Approximately 5 hours are required for compliance with this Service Bulletin.

1.6 **SPECIAL TOOLS**

No special tools are required.

1.7 **VARIATION IN WEIGHT & BALANCE**

Not Applicable

1.8 **ELECTRICAL SYSTEM LOAD**

Not Applicable

1.9 **REFERENCE DOCUMENTS**

Maintenance Manual

1.10 **PUBLICATIONS AFFECTED**

Maintenance Manual

1.11 **APPLICATION TO BE CARRIED OUT BY**

Certified Maintenance Facility.

2. WORK PROCEDURE

- 2.1. Rotate the cockpit fuel selector to all positions, checking for freedom of movement, and check that the detent is perceptible when the knob is placed in each position.

WARNING

Any uncertainty regarding positive detent sensibility may lead to involuntary erroneous positioning of the fuel selector valve.

- 2.2. If the detent is clearly perceptible, perform an operational check as per paragraph 2.4.
- 2.3. In case of an un-perceptible detent, perform trouble shooting as follows:
- 2.3.1 Position the cockpit Fuel Control Knob to the normally open position (Figure 1 refers).
 - 2.3.2 With reference to the Maintenance Manual, remove the wing leading edge inspection panel to gain access to the Fuel Selector Control Valve.
 - 2.3.3 Remove the screws securing the Fuel Control Receiver to its bracket and carefully move the receiver downwards until it disengages. Recover the key.
 - 2.3.4 Insert a 3 mm diameter pin in the hole of the fuel selector control valve shaft. Using the pin as a lever, manually rotate the valve shaft checking for freedom of rotation. In each position the detent signalling valve positioning must be clearly perceptible. **Replace the selector valve if too hard to rotate, or if the detent signalling valve positioning is no longer perceptible. (Refer to the applicable procedure in the Maintenance Manual)**
 - 2.3.5 With the receiver disconnected move the cockpit Control Knob and check for freedom of movement.
 - 2.3.6 If the control knob operates freely (Figures 1 and 2 refer).
 - a. Position the transmitter selector knob and the selector valve to normally open, by observing the markings on the head of the selector valve shaft.
 - b. Reinstall the receiver, positioning the key as per figure 2.
 - c. Close the wing inspection panel.
 - d. Perform an operational check as per paragraph 2.4.
 - 2.3.7 If the control knob is hard to operate (Figures 2, 3 and 4 refer)
 - a. Gain access to the Fuel Control Transmitter removing the cockpit overhead panel.
 - b. Remove the stop tubes from the transmitter and the receiver.
 - c. Slowly rotate the transmitter shaft to extract the cable through the transmitter, and when the cable disengages from the receiver, mark with a reference line the position of the transmitter internal wheel with respect to the body.
 - d. Pull out the cable through the transmitter.
 - e. Disconnect the cable tube from the transmitter and the receiver.
 - f. Place a suitable sized container under the receiver end of the tube to collect solvent and grease which will flow out from the tube during the next step of the procedure.
 - g. Spray methyl-ethyl-ketone solvent in the tube until all the old grease is removed. If necessary, after spraying the solvent, the control cable may be used as a brush taking care not to damage it. Remove the cable and repeat solvent spraying, air dry the tube.

2. WORK PROCEDURE (cont.)

- h. Degrease the cable using methyl-ethyl-ketone solvent and inspect it for condition and wear. **Replace if damaged or worn.**
- i. Manually grease the cable lightly with graphitic grease.
- j. Connect the transmitter and the receiver to the tube.
- k. Install the cable pushing it into the transmitter. Turn the transmitter shaft to help the cable enter the tube checking for freedom of sliding. When the cable impacts the internal wheel of the receiver, position the wheel according to the reference line as indicated in step “c”. Continue pushing the cable to engage the receiver. When pushed the cable may be rotated counter-clockwise to facilitate engagement of the receiver.

NOTE: In case of stiffness check the cable tube for deformation or dents.
A damaged tube that obstructs the cable from sliding freely must be replaced or repaired. If necessary, contact VulcanAir for further instructions.

- l. Position the selector valve to normally open, by observing the markings on the head of the valve shaft. Refer to Figure 1.
- m. Turn the transmitter shaft until $2.06 \pm 1/8$ inches of the cable protrudes from the transmitter.
- n. Install the receiver by aligning the two key slots on the receiver and the valve shaft respectively. Check that the protrusion of the cable is in accordance with the values indicated in step “m”. If necessary, pull back the cable to disengage the receiver and repeat step “k” adjusting the receiver internal wheel position to obtain correct rigging.
- o. Install the receiver, positioning the key as indicated in Figure 2.
- p. **For aircraft up to S/N 122**, install the receiver and transmitter stop tubes. Rotate the cockpit fuel selector control knob in all its positions and check that the markings on the head of the valve shaft are in accordance with Figure 1.
- q. **For aircraft from S/N 123 onwards**, rotate the cockpit fuel selector control knob to the ENG SHUT OFF position. Install the receiver stop tube as indicated in Figure 4 and before tightening the cable stop screw, check that the marking on the head of the valve shaft is set as indicated in Figure 1 (ENG. SHUT OFF). Tighten the stop screw at 2 mm from the extremity of the slot in the stop tube as indicated in detail A of Figure 4. Rotate the cockpit fuel selector control knob in all its positions and check that the markings on the head of the valve shaft are in accordance with Figure 1. Install transmitter stop tube.
- r. Install the overhead panel and all previously removed items.
- s. Position the cockpit fuel selector control knob to normally open.
- t. Install the fuel transmitter control knob with its pointer centered on the normally open position.
- u. Close wing inspection panel.
- v. Perform an operational check as per paragraph 2.4.

2. WORK PROCEDURE (cont.)

2.4 FUEL SYSTEM OPERATIONAL CHECK

2.4.1 Tow the aircraft to a suitable engine test area.

2.4.2 Ensure that the quantity of fuel in the tanks is enough to guarantee at least thirty minutes of continuous engine operation.

2.4.3 Position the aircraft upwind, and place chocks under the main landing gear wheels.

2.4.4 Start both engines according to the procedure in the relative Flight Manual.

2.4.5 Run both engines at 1,500 RPM.

2.4.6 Place the left engine fuel selector control knob in the ENG SHUT OFF position and verify that the LH engine shuts down. **If engine does not shut down, replace the fuel selector control valve.**

2.4.7 Place the left engine fuel selector control in the LH TANK position.

2.4.8 Restart the LH engine and run at 1,500 RPM.

2.4.9 **For Aircraft from Serial Number 123 onwards:** Place the left engine fuel selector control knob in the ENG SHUT OFF position, and when it begins to shut down, place the selector control knob in the RIGHT TANK position and verify that the engine runs normally. If necessary turn on the electrical FUEL PUMP relative to the tank in use. Allow engine to run in this regime for at least 2 minutes.

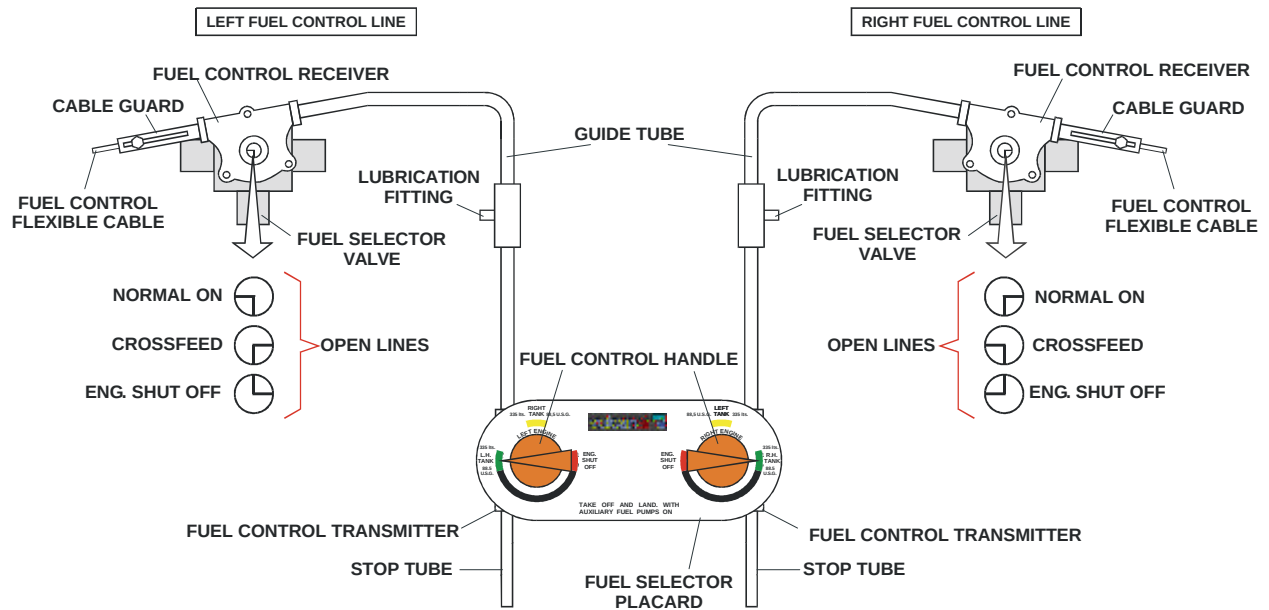
2.4.10 **For Aircraft up to Serial Number 122:** Place the left engine fuel selector control knob in the ENG SHUT OFF position and when it begins to shut down, place the left selector control knob in the TANK OFF position and the right selector control knob in the CROSS FEED position. Verify that the engine runs normally. If necessary turn on the electrical FUEL PUMP relative to the tank in use. Allow engine to run in this regime for at least 2 minutes.

2.4.11 Repeat the applicable procedure above for the RH engine, inverting the selections as required.

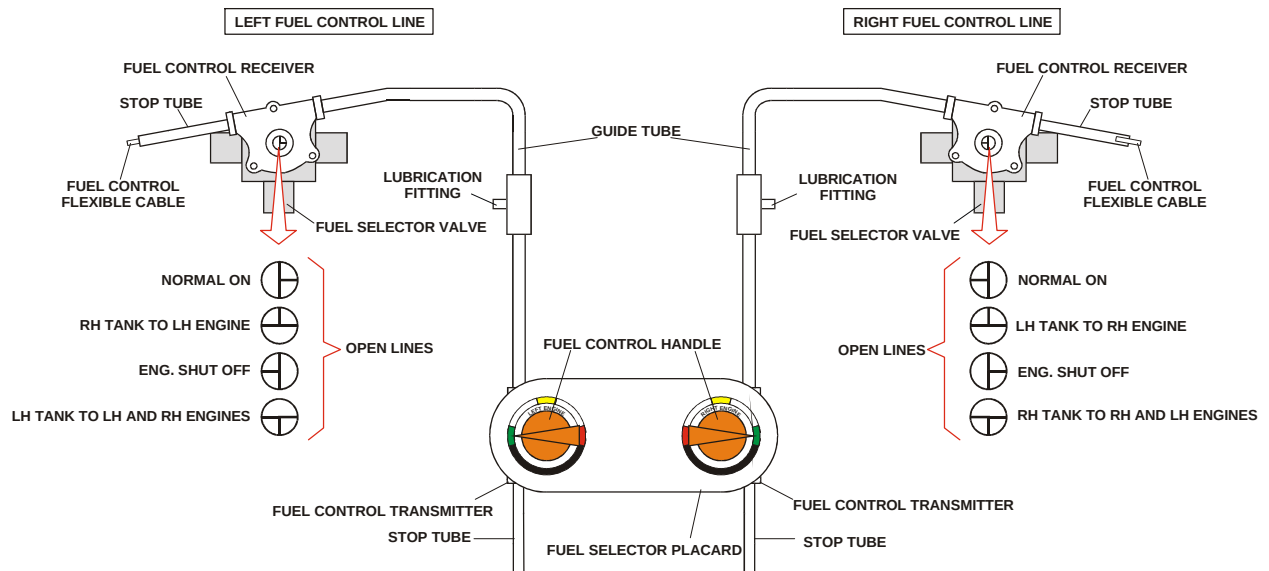
2.4.12 Insert this Service Bulletin in the applicable Section of the Maintenance Manual until a new revision is issued.

2.4.13 Record compliance with this Service Bulletin in the aircraft Log-Book.

2.4.14 **To ensure crewmembers are aware of the contents of the present Bulletin, insert a copy of this Bulletin in the Aircraft Flight Manual applicable section.**



For P68 Series Aircraft from Serial Number 123 onwards



For P68 Series Aircraft up to Serial Number 122

Figure 1 – Fuel Control System Scheme

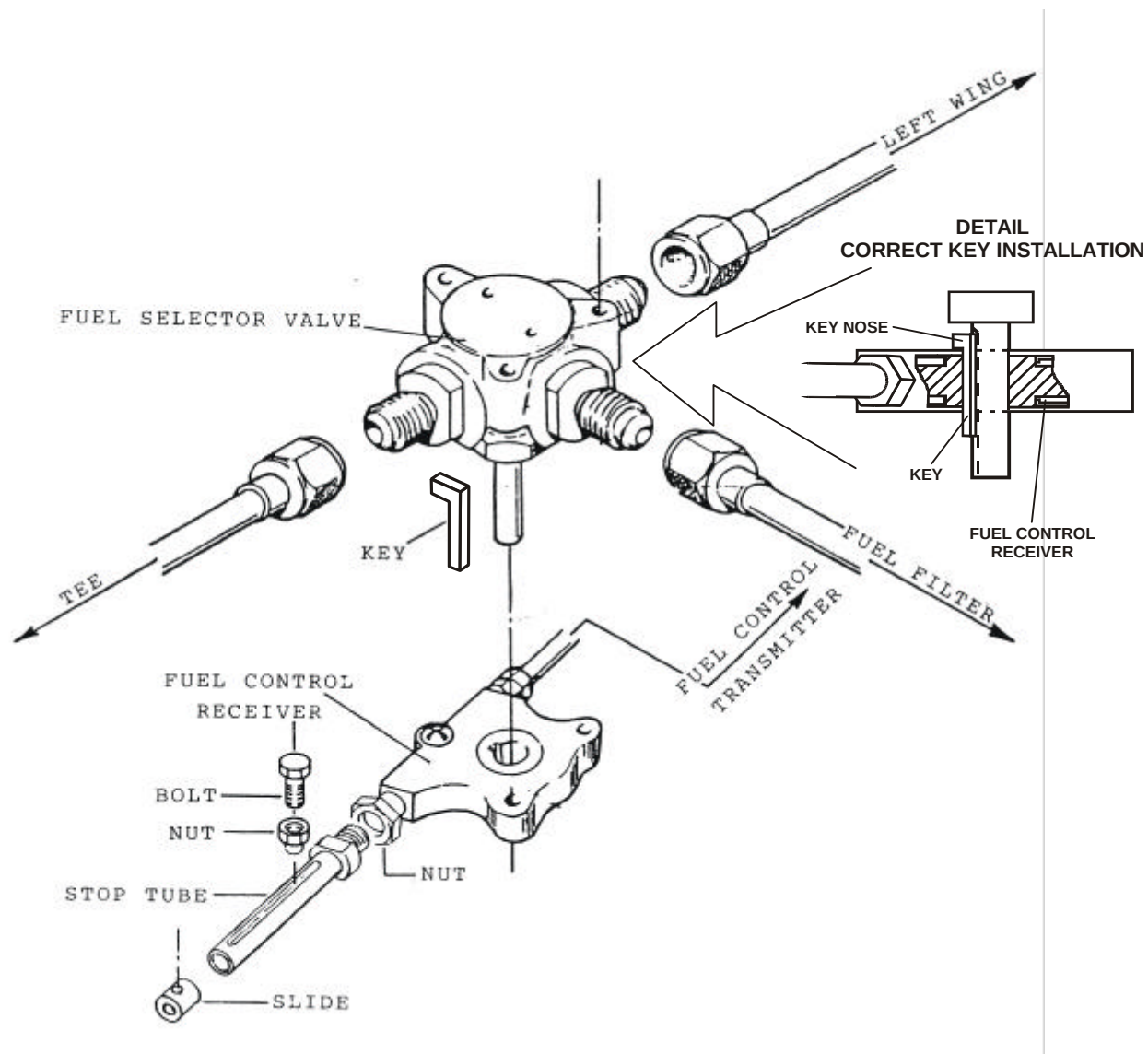


Figure 2 – Fuel Selector Valve and Receiver Installation

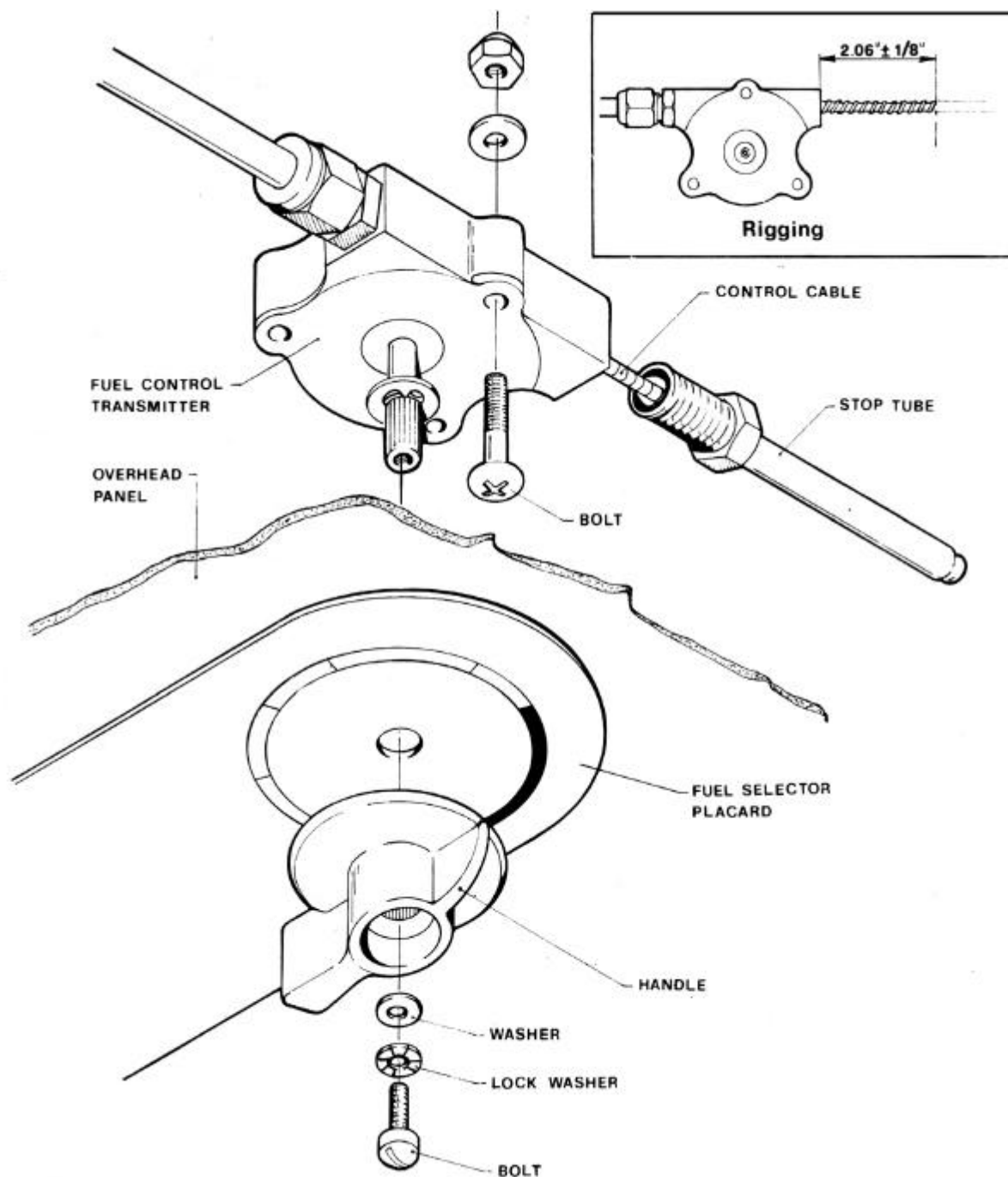


Figure 3 – Cockpit Fuel Selector Control Installation

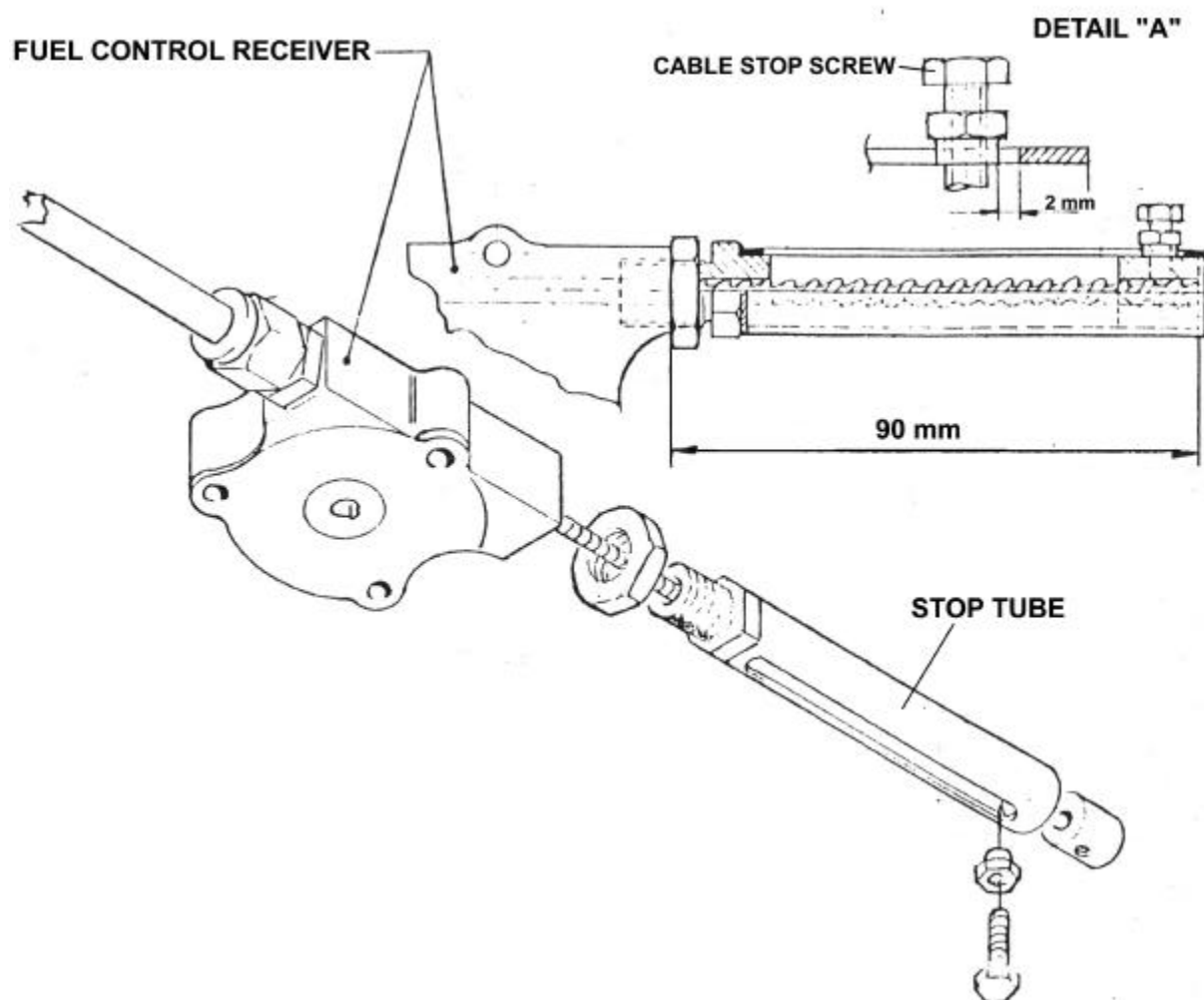


Figure 4 – Fuel Selector Control Receiver for P68 aircraft from Serial Number 123 onwards